Multi-dimensional instability in energy financial markets: A delayed asymmetric agent-based modeling approach

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Abstract

Global financial markets frequently experience severe asset price volatility and recurrent extreme risk events, posing significant threats to financial stability. Combining financial physics and agent theory perspectives, this study develops a Multi-dimensional Instability Index (MDI) along with a delayed asymmetric agent-based model to systematically characterize market instability. The proposed MDI comprehensively integrates multiple key indicators, including information entropy, unstable time, absolute return volatility, maximum drawdown, and value-at-risk (VaR). Empirically analyzing WTI crude oil futures from June 1988 to October 2024, the paper provides in-depth insights into market stability dynamics during notable crises such as the 2008 financial crisis, the 2014–2016 oil price collapse, the COVID-19 pandemic, and recent geopolitical risks. Simulation results reveal a nonlinear, non-monotonic relationship between trading probabilities and market stability, identifying optimal investor behaviors that enhance stability. Furthermore, changes in information delay, behavioral asymmetry, and agent clustering significantly influence market stability, exhibiting bifurcation dynamics. Findings highlight the necessity of improving information dissemination efficiency and reducing investor behavioral asymmetry to strengthen financial stability, offering valuable quantitative tools and theoretical guidance for macroprudential regulation and systemic risk early-warning systems.

Keywords: Econophysics, Financial Crisis, Market Stability, Multi-dimensional Instability Index, Delayed Asymmetric Agent-Based Model, Systemic Risk Management

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